UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/123,208	03/17/2023	Yu Fang	SONA-0010-U01-C01-C12	4903

87084 7590 10/05/2023

GTC Law Group PC & Affiliates
One University Ave., Ste. 302B
Westwood, MA 02090

EXAMINER	
ALGIBHAH, HAMZA N	

ART UNIT	PAPER NUMBER
2456	

NOTIFICATION DATE	DELIVERY MODE
10/05/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 18/123,208	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2456	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 03/17/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-5 and 8-20 is/are rejected.

8) ☑ Claim(s) 6-7 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 03/17/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims 1-20*** are pending.

2. ***Claims* 1-5 and 8-20 *are rejected.***

3. ***Claims 6-7 are objected.***

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

4. ***Claim***(s) **1-5 and 8-20** is/are rejected under 35 U.S.C. 103 as being unpatentable over Duri et al (Pub. No.: US 2004/0267410 A1) in view of Thakur et al (Pub. No.: US 2019/0007794 A1).

As per claim 1, Duri discloses a ~~cloud~~ system, comprising:

- a request interface (Duri, Fig 4 item 425) configured to interpret a vehicle data collection request for at least one identified vehicle (Duri, paragraph 0074, wherein a processor executing in the protected environment can monitor for received events and requests for telematics data. As noted, events can be received from the vehicle indicating a need for a particular service. Requests for telematics data can be received from external applications and/or trusted agents); and

- a policy creator circuit (Duri, Fig 4) structured to determine a data collection policy in response to the vehicle data collection request (Duri, paragraph 0076, wherein the relevant privacy policy can be identified and retrieved. More particularly, the privacy policy corresponding to the subject vehicle, or source of the telematics data, can be retrieved. In the

case where an event is received, the vehicle source of the event can be identified from the event and a corresponding privacy policy can be retrieved. If a request for telematics information is received, then either a specific vehicle can be identified from the request for information, or more than one vehicle can be identified from the request);

- ***a ~~cloud~~ interface*** (Duri, Fig 4) ***configured to provide the data collection policy to the at least one identified vehicle*** (Duri, paragraph 0076, wherein the relevant privacy policy can be identified and retrieved. More particularly, the privacy policy corresponding to the subject vehicle, or source of the telematics data, can be retrieved. In the case where an event is received, the vehicle source of the event can be identified from the event and a corresponding privacy policy can be retrieved. If a request for telematics information is received, then either a specific vehicle can be identified from the request for information, or more than one vehicle can be identified from the request)***, and to receive responsive vehicle data from the at least one identified vehicle*** (Duri, paragraph 0073, wherein telematics data can be collected from sensors within the vehicle as well as received from the communications bus of the vehicle. The telematics data can be received at periodic intervals, from time to time, or can be received when particular items of data change. Regardless, the telematics data can be continually received and updated)***; and***

- ***a raw data manager circuit structured to store at least a portion of the responsive vehicle data*** (Duri, paragraph 0016, 0034, wherein the system can include a data store having telematics data for the vehicle and a data store having privacy policy information corresponding to the vehicle and an application service provider. The TSP 140 can receive telematics data and events from vehicle 105 via wireless transmissions 135 and store the received data). Duri does not explicitly disclose that the system is ***a cloud system comprising a cloud interface***. However, Thakur discloses that the system is ***a cloud system comprising a cloud interface*** (Thakur, Fig 2,

paragraph 0022, 0042 wherein the cloud platform 202 can include a plurality of components including but not limited to a third-party service API, application logic, and vehicle interaction API).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify Duri in view of Thakur such that the system is a cloud system as claimed because this would have provided a way to improve the system performance by using the power of a cloud system which provide a much faster hardware structure.

As pre claim 2, claim 1 is incorporated and Thakur discloses **wherein the request interface is further configured to expose an application programming interface (API) to an external device, and to interpret the vehicle data collection request in response to an exercise of the API** (Thakur, Fig 2, paragraph 0042, 0045 wherein cloud platform 202 can facilitate communications (e.g., sending messages, alerts, triggering actions, etc.) between the target vehicle 204 and the delivery vehicle 206 using the vehicle interaction API and can communicate with both the target vehicle 204 and the delivery vehicle 206. For example, the cloud platform 202 can receive data communications from the target vehicle 204 regarding vehicle status indications that the target vehicle 204 (or an onboard device such as a TCU) has detected (e.g., fuel level low) and the cloud platform 202 can send additional information or determinations made based on the received data communications to the delivery vehicle 206 (e.g., a refueling on-demand service need alert or a location of the target vehicle 204 so that the delivery vehicle 206 automatically navigates to the location if available). In addition, providing access to and securing the target vehicle 204 can be facilitated by an instruction sent by the cloud platform

202 or by the smartphone 208 that triggers the locking/unlocking mechanisms via the CAN bus of the target vehicle 204);

As pre claim 3, claim 2 is incorporated and Thakur discloses **wherein the request interface is further configured to interpret a vehicle data request from the external device in response to an exercise of the API, to retrieve at least a portion of the responsive vehicle data from the raw data manager circuit in response to the vehicle data request, and to provide the retrieved vehicle data to the external device** (Thakur, Fig 2, paragraph 0042, 0045 wherein cloud platform 202 can facilitate communications (e.g., sending messages, alerts, triggering actions, etc.) between the target vehicle 204 and the delivery vehicle 206 using the vehicle interaction API and can communicate with both the target vehicle 204 and the delivery vehicle 206. For example, the cloud platform 202 can receive data communications from the target vehicle 204 regarding vehicle status indications that the target vehicle 204 (or an onboard device such as a TCU) has detected (e.g., fuel level low) and the cloud platform 202 can send additional information or determinations made based on the received data communications to the delivery vehicle 206 (e.g., a refueling on-demand service need alert or a location of the target vehicle 204 so that the delivery vehicle 206 automatically navigates to the location if available). In addition, providing access to and securing the target vehicle 204 can be facilitated by an instruction sent by the cloud platform 202 or by the smartphone 208 that triggers the locking/unlocking mechanisms via the CAN bus of the target vehicle 204);

As pre claim 4, claim 1 is incorporated and Duri discloses **wherein the request interface is further configured to interpret a vehicle data request from an external device** (Duri, paragraph 0074, wherein a processor executing in the protected environment can monitor for

received events and requests for telematics data. As noted, events can be received from the vehicle indicating a need for a particular service. Requests for telematics data can be received from external applications and/or trusted agents}**, to retrieve at least a portion of the responsive vehicle data from the raw data manager circuit in response to the vehicle data request** (Duri, paragraph 0077, wherein the data to be provided to the requestor can be determined as specified by the retrieved privacy policy. The privacy policy can be compared with the stored telematics data and any received events. It should be appreciated that while the telematics data is dynamic in nature, items of telematics information representing an instance or moment in time can be compared with the privacy policy)**, and to provide the retrieved vehicle data to the external device** (Duri, paragraph 0016, 0034, wherein the system can include a data store having telematics data for the vehicle and a data store having privacy policy information corresponding to the vehicle and an application service provider. The TSP 140 can receive telematics data and events from vehicle 105 via wireless transmissions 135 and store the received data);

As pre claim 5, claim 1 is incorporated and Duri discloses wherein the request interface is configured to interpret at least one additional vehicle data collection request for the at least one identified vehicle (Duri, paragraph 0074, wherein a processor executing in the protected environment can monitor for received events and requests for telematics data. As noted, events can be received from the vehicle indicating a need for a particular service. Requests for telematics data can be received from external applications and/or trusted agents}, and wherein the policy creator circuit is further structured to determine the data collection policy in response to the vehicle data collection request and the at least one additional vehicle data collection request (Duri, paragraph 0076, wherein the relevant privacy policy can be identified

and retrieved. More particularly, the privacy policy corresponding to the subject vehicle, or source of the telematics data, can be retrieved. In the case where an event is received, the vehicle source of the event can be identified from the event and a corresponding privacy policy can be retrieved. If a request for telematics information is received, then either a specific vehicle can be identified from the request for information, or more than one vehicle can be identified from the request);

As pre claim 8, claim 1 is incorporated and Duri discloses wherein the policy creator circuit is further structured to determine a policy capability value in response to the vehicle data collection request, and to selectively enable determining the data collection policy in response to the policy capability value (Duri, paragraph 0012, 0028 wherein the method can include receiving the telematics data for the vehicle, wherein the telematics data dynamically changes over time, and comparing the telematics data with a privacy policy associated with the vehicle. The privacy policy can specify conditions for selectively releasing items of the telematics data to one or more service providers. In particular, the present invention allows one to define a privacy policy that can specify which items of telematics information received from a vehicle are to be shared with service providers and the conditions under which those items are to be shared; wherein the value that is used to determine what data to transmit can be the transmission capability value as claimed);

As pre claim 9, claim 8 is incorporated and Duri discloses wherein the policy creator circuit is further structured to determine the policy capability value in response to at least one parameter selected from the parameters consisting of a data storage size determined to support the vehicle data collection request; a transmission amount value determined to

support the vehicle data collection request; a data availability value associated with the vehicle data collection request; or a data configuration value associated with the vehicle data collection request (Duri, paragraph 0009, 0078 wherein each service provider also can have a corresponding privacy policy. In that case, a received request for telematics data can specify the service provider's privacy policy, whether the privacy policy is provided with the request or is accessed after receiving the request, for example from a local or remote data store. The data owner's privacy policy can be compared with the service provider's privacy policy to determine whether the two policies can be reconciled. For example, the service provider's privacy policy can specify how a data owner's telematics data (and other information) will be handled after it is acquired. If the service provider's privacy policy cannot be reconciled with the data owner's privacy policy, no information need be provided to the service provider. Alternatively, only telematics data for which both privacy policies can be reconciled can be provided);

As pre claim 10, claim 1 is incorporated and Duri discloses wherein the policy creator circuit is further structured to determine a policy authorization value in response to the vehicle data collection request, and to perform at least one operation, in response to the policy authorization value, selected from the operations consisting of: selectively enabling the determining the data collection policy; or determining the data collection policy to support at least a portion of the vehicle data collection request (Duri, paragraph 0043, 0047 wherein the PERM 215 can query the PAD 210 for authorization to respond to received data requests. After receiving authorization, the PERM 215 can read data from the profile data store 225.and a corresponding privacy policy in the policy data store 230. The PERM 215 then can return information complying with the consulted privacy policy to the requestor via the responder 205.

For example, according to one embodiment of the present invention, the PERM 215 can first obtain authorization from the PAD 210 prior to accessing information. In that case, the PERM 215 can be restricted in that only the information that the PERM 215 is authorized to access is retrieved or read. Alternatively, the PERM 215 can be configured to retrieve all data pertaining to a vehicle. Accordingly, the PERM 215 can submit the retrieved or read data from the profile data store 225 to the PAD 210. The PAD 210 then can remove any data that the requester is not authorized to receive);

As pre claim 11, claim 1 is incorporated and Duri discloses wherein the request interface is configured to provide at least one use case value to a user interface, each use case value comprising a vehicle data collection template, and determining the vehicle data collection request in response to responses from the user interface to the provided at least one use case value (Duri, paragraph 0076, wherein If a request for telematics information is received, then either a specific vehicle can be identified from the request for information, or more than one vehicle can be identified from the request. For example, while an ASP can request information corresponding to a single vehicle, the situation also can arise where an ASP requests information for each vehicle within a particular geographic region, for particular vehicles passing through a geographic region within a particular time frame, for all vehicles belonging to a class of vehicles, etc.);

As pre claim 12, claim 11 is incorporated and Beiderbeck discloses wherein the request interface is further configured to determine the at least one use case value in response to at least one of: an entity type associated with the user interface; a permissions value associated with the user interface; and previous data collection policies determined for users having a

shared characteristic determined for the user interface (Duri, paragraph 0076, wherein If a request for telematics information is received, then either a specific vehicle can be identified from the request for information, or more than one vehicle can be identified from the request. For example, while an ASP can request information corresponding to a single vehicle, the situation also can arise where an ASP requests information for each vehicle within a particular geographic region, for particular vehicles passing through a geographic region within a particular time frame, for all vehicles belonging to a class of vehicles, etc.);

As pre claim 13, claim 1 is incorporated and Beiderbeck discloses wherein the at least one identified vehicle comprises a group of vehicles (Duri, paragraph 0076, wherein If a request for telematics information is received, then either a specific vehicle can be identified from the request for information, or more than one vehicle can be identified from the request. For example, while an ASP can request information corresponding to a single vehicle, the situation also can arise where an ASP requests information for each vehicle within a particular geographic region, for particular vehicles passing through a geographic region within a particular time frame, for all vehicles belonging to a class of vehicles, etc.);

As pre claim 14, claim 13 is incorporated and Beiderbeck discloses wherein the group of vehicles comprises a fleet of vehicles owned by a common entity (Duri, paragraph 0076, wherein If a request for telematics information is received, then either a specific vehicle can be identified from the request for information, or more than one vehicle can be identified from the request. For example, while an ASP can request information corresponding to a single vehicle, the situation also can arise where an ASP requests information for each vehicle within a

particular geographic region, for particular vehicles passing through a geographic region within a particular time frame, for all vehicles belonging to a class of vehicles, etc.);

As pre claim 15, claim 1 is incorporated and Duri discloses a requestor verification circuit structured to: interpret a property request value from a requesting entity, the requesting entity comprising at least one of an end point, a flow, or an application of the at least one identified vehicle; and determine an entity authorization value corresponding to the end point, flow, or application; and a vehicle property provisioning circuit structured to transmit the stored at least a portion of the responsive vehicle data to the requesting entity (Duri, paragraph 0043, 0047 wherein the PERM 215 can query the PAD 210 for authorization to respond to received data requests. After receiving authorization, the PERM 215 can read data from the profile data store 225.and a corresponding privacy policy in the policy data store 230. The PERM 215 then can return information complying with the consulted privacy policy to the requestor via the responder 205. For example, according to one embodiment of the present invention, the PERM 215 can first obtain authorization from the PAD 210 prior to accessing information. In that case, the PERM 215 can be restricted in that only the information that the PERM 215 is authorized to access is retrieved or read. Alternatively, the PERM 215 can be configured to retrieve all data pertaining to a vehicle. Accordingly, the PERM 215 can submit the retrieved or read data from the profile data store 225 to the PAD 210. The PAD 210 then can remove any data that the requester is not authorized to receive);

As pre claim 16, claim 15 is incorporated and Duri discloses a subscription circuit structured to add the requesting entity to a subscriber data list in response to the property request value and the entity authorization value; and wherein the vehicle property

provisioning circuit is further structured to transmit the stored at least a portion of the responsive vehicle data to the requesting entity in response to the subscriber data list (Duri, paragraph 0037, 0042, 0059, wherein for example, ASP 155 can subscribe or receive information for vehicles within a particular geographic region, as was the case with ASP 150. In addition to supporting request/response protocol, the PERM 215 also can support the subscription model for data. The notification dispatcher 465 can generate notifications to be posted to the request processor 425 which can then be forwarded to subscribers);

Claims 17-20 are rejected under the same rationale as claims 1-5 and 8-16.

Allowable Subject Matter

5. Claims 6-7 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached on 7:30 am - 3:30 pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Umar Cheema can be reached on (571) 270-3037. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2456